

Procedimiento algebraico Funcion de transferencia

$$\frac{V_2(s)}{V_e(s)} = \frac{(\text{polinomio}(s)) I_2(s)}{(P_s^2 + P_s + ?) I_2(s)}$$

1. Agrupar corrientes $I_1(s)$ e $I_2(s)$

2. Expresar a $V_e(s)$ en función de $I_2(s)$.

3. Despejar $I_1(s)$ de la segunda ecuación principal.

$$V_2(s) = (R + \frac{1}{Cs}) I_2(s) = (\frac{CRs + 1}{Cs}) I_2(s)$$

$$Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$Ls I_1(s) + R I_1(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs} + Ls I_2(s) + R I_2(s)$$

$$(Ls + R) I_1(s) = (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_e(s) = R I_2(s) + Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) - (Ls + R) I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} - (Ls + R) \right] I_2(s)$$

$$V_e(s) = \int \frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLRs^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)}$$

$$= \frac{\frac{CRs + 1}{Cs}}{\frac{3CLRs^2 + (5CR^2 + L)s + 2R}{Cs(Ls + R)}}$$

$$= \frac{(Ls + R)(CRs + 1)}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

Error en estado estacionario cuando se aplica un escalón unitario de entrada

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{C L R s^2 + (C R^2 + L) s + R}{3 C L R s^2 + (5 C R^2 + L) s + 2 R} \right) = \frac{R}{2 R}$$

$$e_t = 0.5 r$$

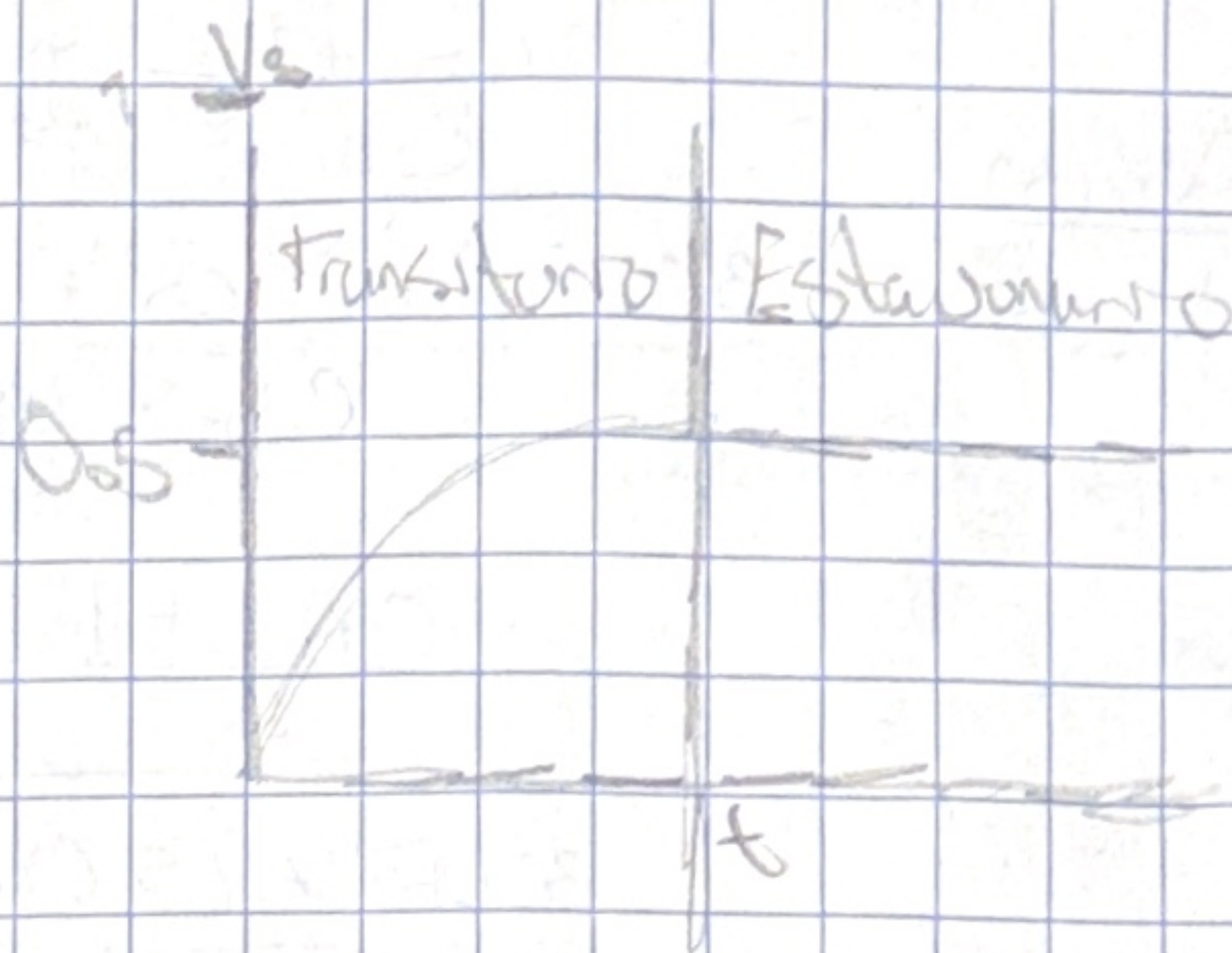
Estabilidad en lazo abierto

Se calcula los polos del sistema (Raíces del denominador).

$$R = 40.7 \text{ k}\Omega$$

$$L = 1.5 \text{ mH}$$

$$C = 470 \mu\text{F}$$



$$3 C L R s^2 + (5 C R^2 + L) s + 2 R = 0$$

$$\begin{matrix} 9.9405 \times 10^{-3} & 51911.5015 & 9400 \\ a & b & c \end{matrix}$$

$$-51911.5015 \pm \sqrt{(51911.5015)^2 - 4(9.9405 \times 10^{-3})(9400)} \\ 2(9.9405 \times 10^{-3})$$

$$-51911.5015 \pm \sqrt{2694803988 - 373.7628}$$

$$-51911.5015 \pm 51911.4979$$

$$0.019881$$

$$X = -0.1659876264$$

$$X = -5695035.295$$